

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒	Optical	☐	Mechanical
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	16 Mar 2017	First Release	MUHAMMED SHAMEEM

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

Appendix A:.....	3
A. Optical PIP Sensor 2 Removal.....	4
I. Optical PIP Sensor 2 Emitter Removal.....	4
II. Optical PIP Sensor 2 Receiver Removal.....	9
B. Installation of the Optical PIP Sensor 2 (Part ID: 4XASP-A151 / 2000-0016).....	12
I. Installation of the Sensor 2 Emitter.....	12
II. Installation of the Sensor 2 Receiver.....	19

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

Appendix A:

1. Locate identical Optical PIP Sensor by refer to Figure 1.

Remarks:Loading sequence: Left to Right.

2. Below is the list and the part number for each of the Optical PIP Kit:

4XASP-A150 / 2000-0015 : S2EX Left Stop PIP Cable Kit (Sensor 1 Emitter and Receiver)

4XASP-A151 / 2000-0016 : S2EX Left Slow PIP Cable Kit (Sensor 2 Emitter and Receiver)

4XASP-A153 / 2000-0017 : S2EX Right Slow PIP Cable Kit (Sensor 3 Emitter and Receiver)

4XASP-A152 / 2000-0018 : S2EX Right Stop PIP Cable Kit (Sensor 4 Emitter and Receiver)

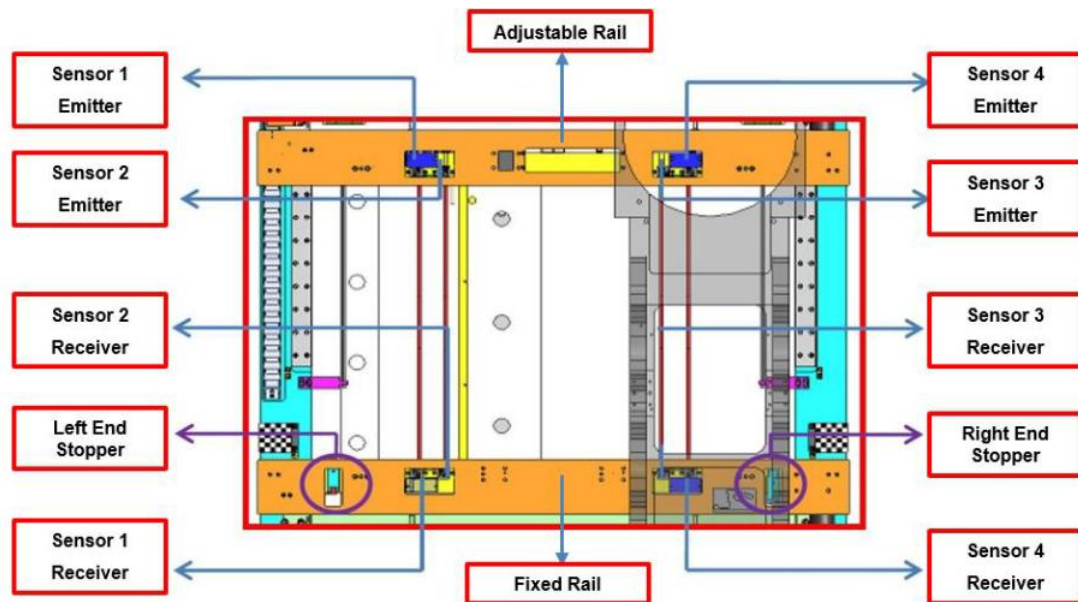


Figure 1 Optical PIP Sensor Location (Graphic)

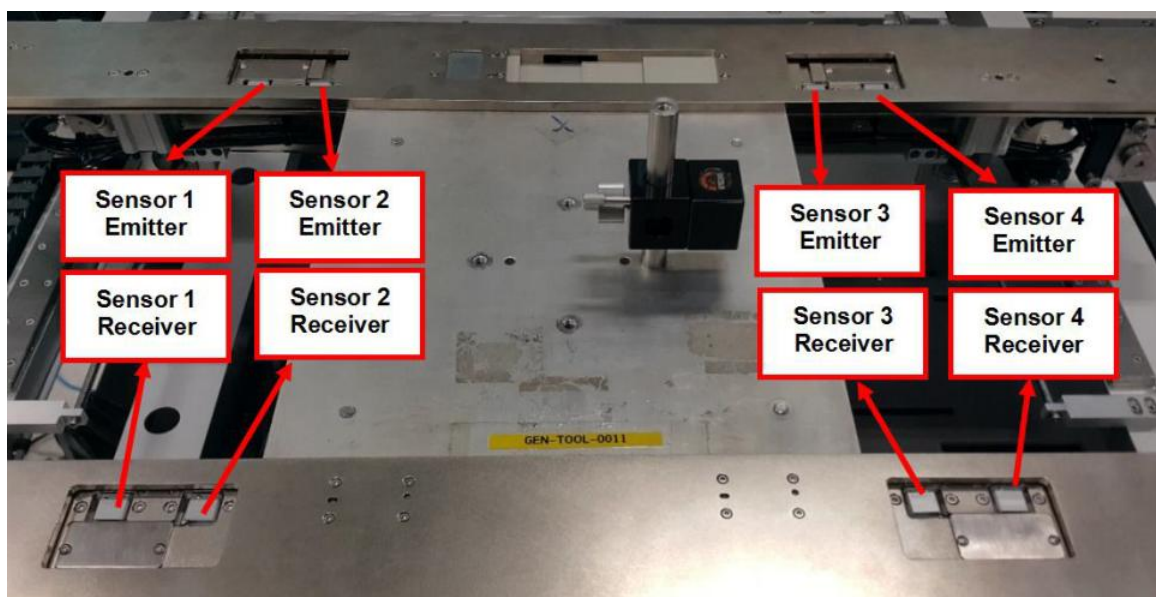


Figure 2 Optical PIP Sensor Location (Physical)

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

A. Optical PIP Sensor 2 Removal

I. Optical PIP Sensor 2 Emitter Removal

1. Open rear access door as Figure 3.

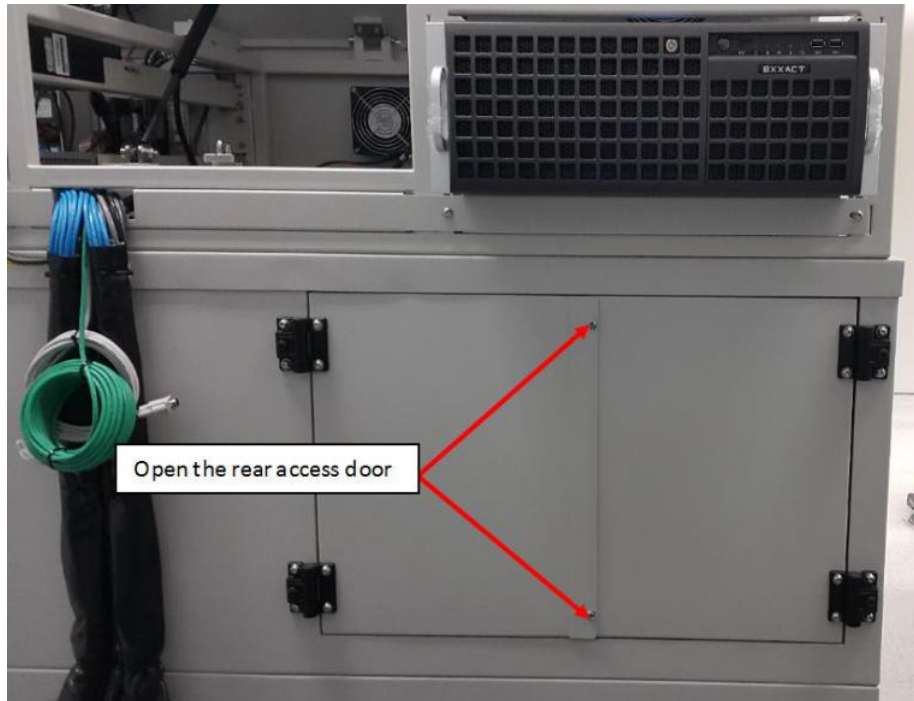


Figure 3 S2EX Rear Door

2. Open BHCS M3x6 (2x) as shown in Figure 4 at adjustable rail (Near to Left Outer Barrier)

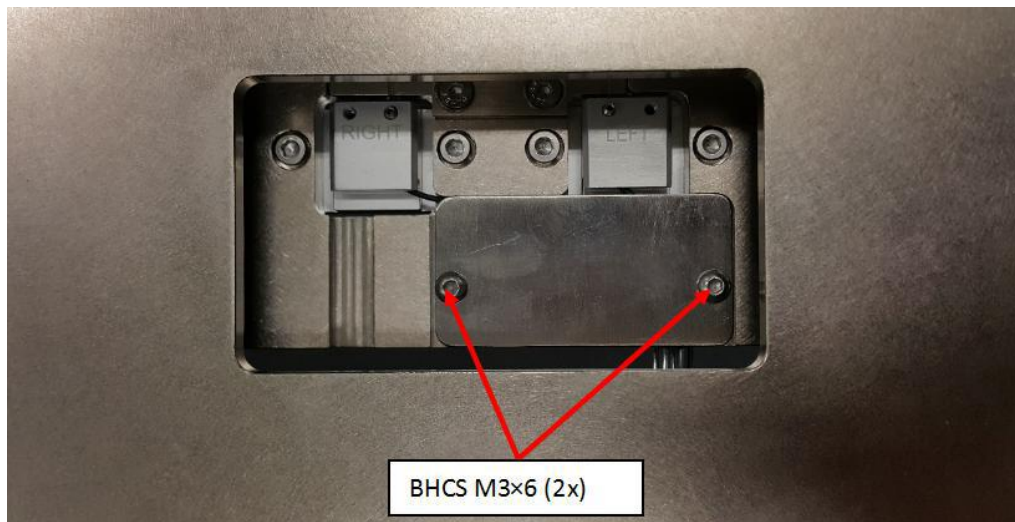


Figure 4 Sensor 2 Emitter

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

3. Open SHCS M3×6 (2x) from Sensor 2 Emitter as shown in Figure 5. Then loosen SHCS M2.5×3. Remove the Optical Sensor 2 Emitter head from pip sensor mounting plate as shown in Figure 6

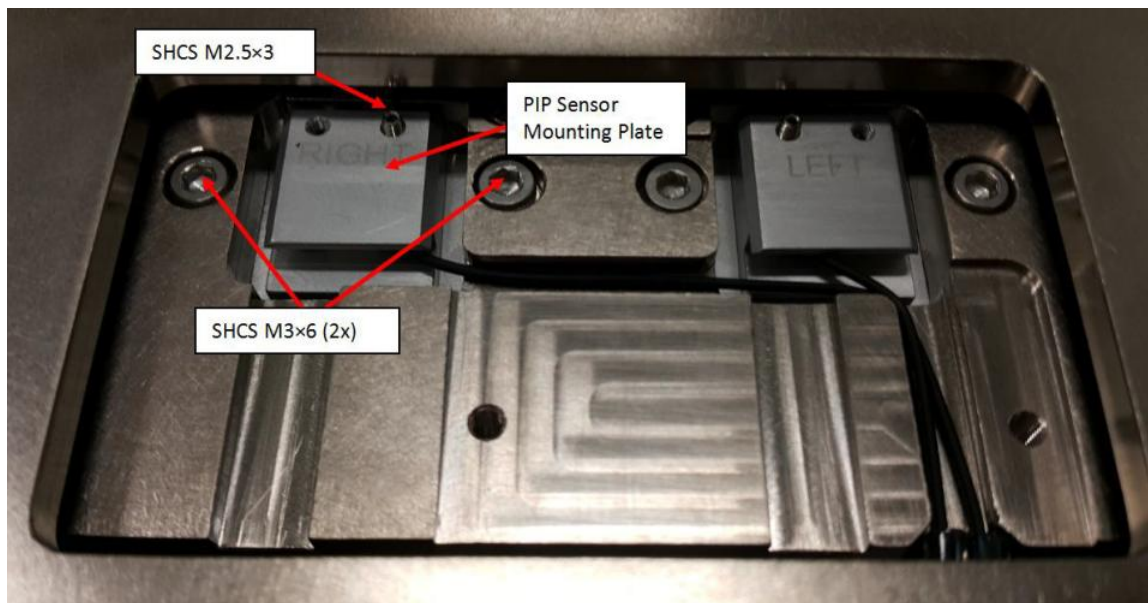


Figure 5 Sensor 2 Emitter

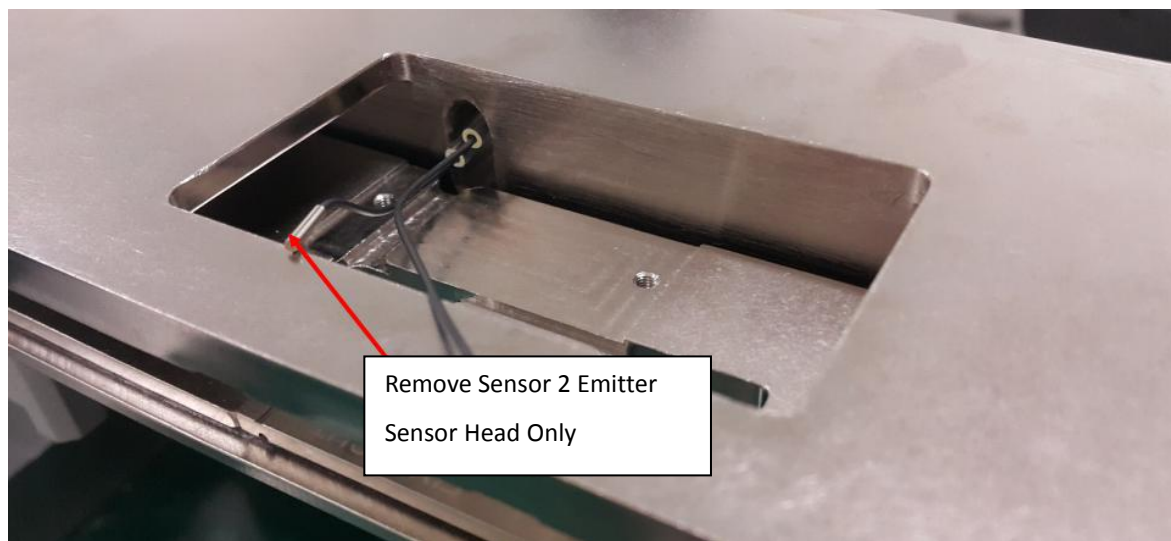


Figure 6 Sensor 2 Emitter (Sensor Head)

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

4. Remove SHCS M4×12 (2x) and cable clamp as shown below in Figure 7. Remove entire Sensor 2 Emitter Head cable as shown from Figure 7 to Figure 13.

Remarks: Do remove any cable ties which are necessary in Sensor 2 Emitter removal.



Figure 7 SHCS M4x12

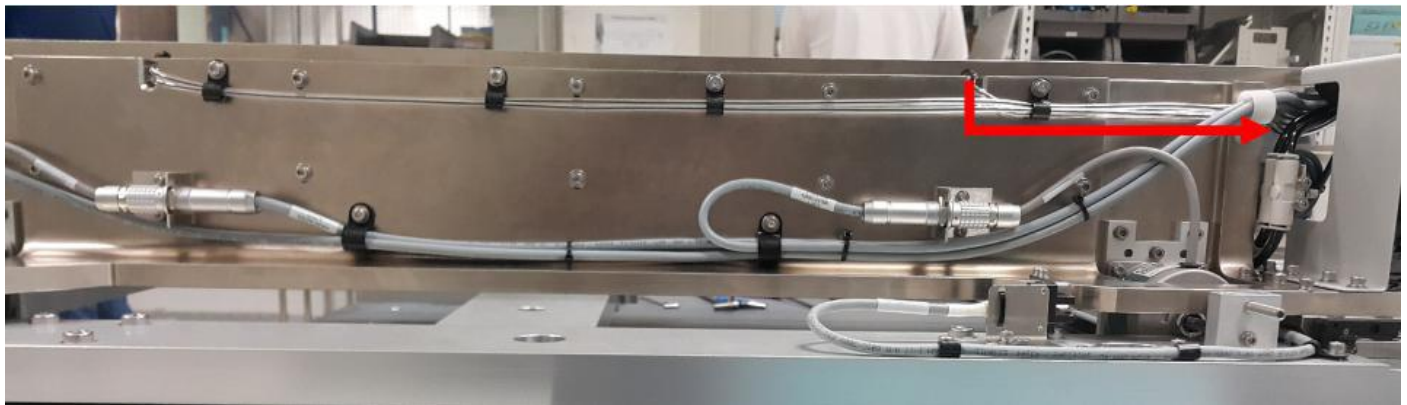


Figure 8 Sensor 2 Emitter Cable Routing

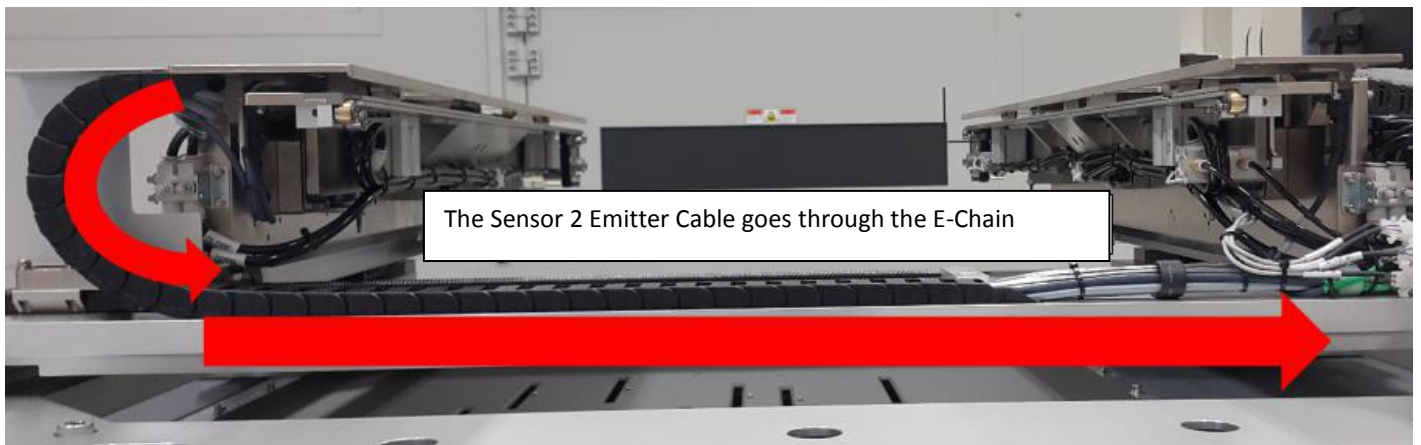


Figure 9 Sensor 2 Emitter (AWA E-Chain)

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

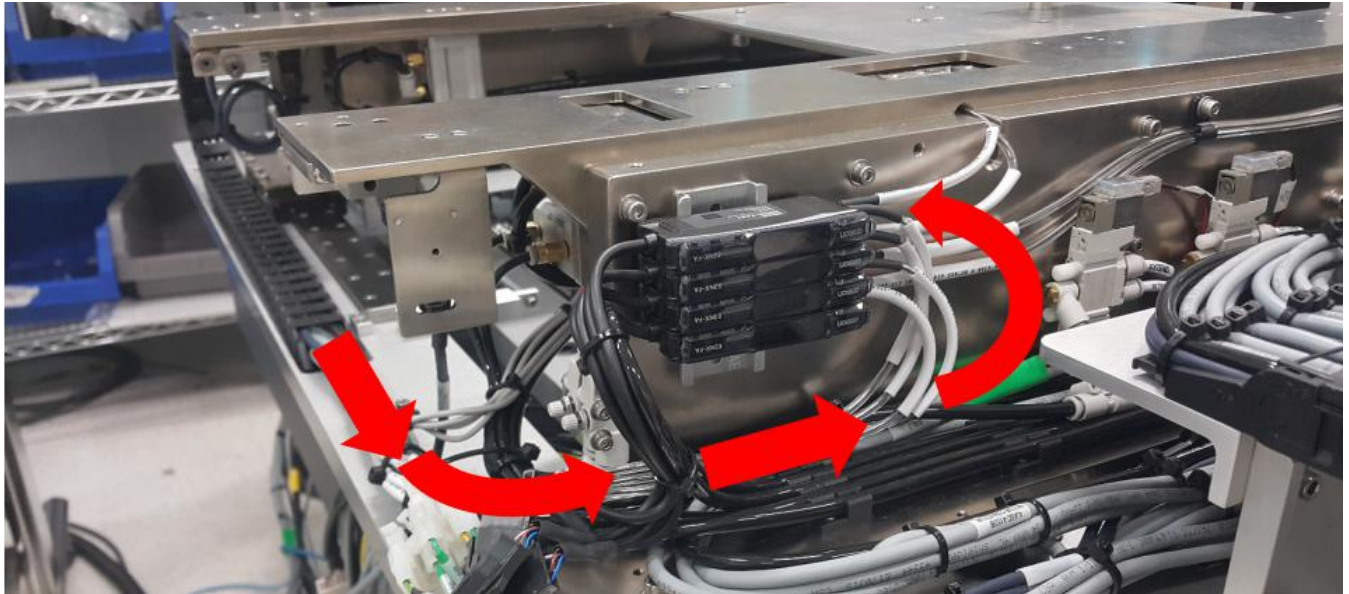


Figure 10 Sensor 2 Emitter (Sensor Amplifier)

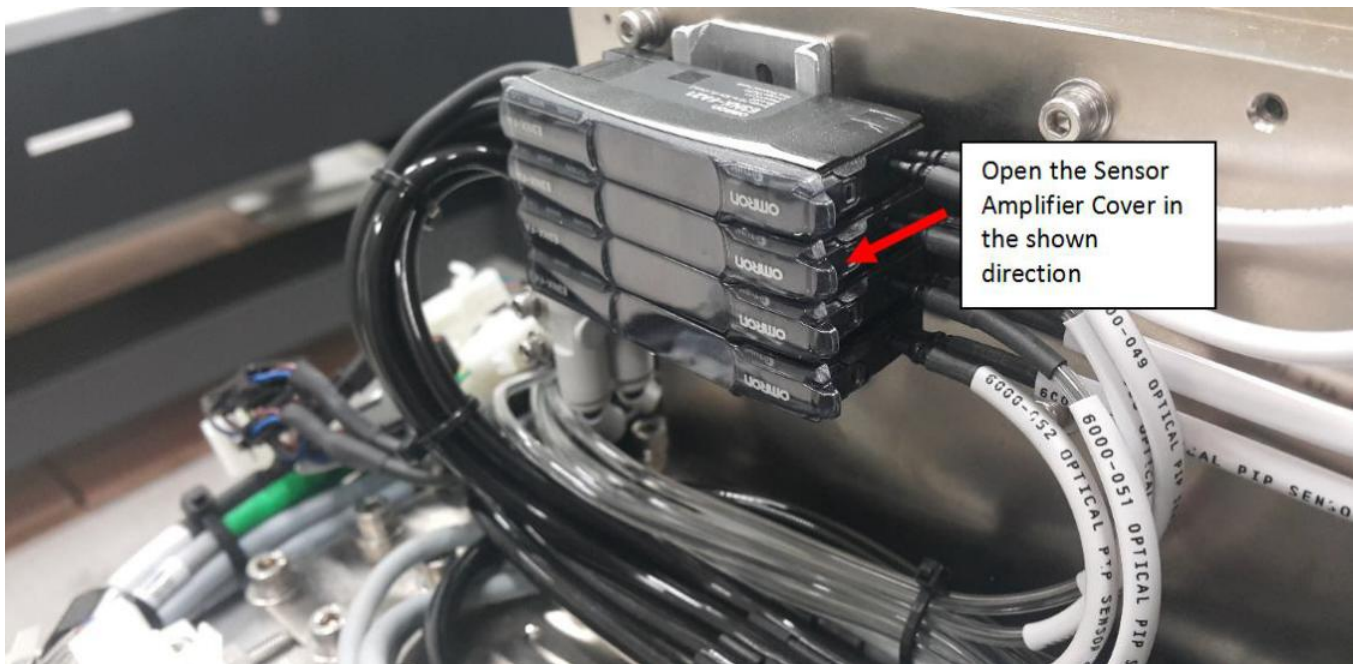


Figure 11 Sensor Amplifier Cover

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

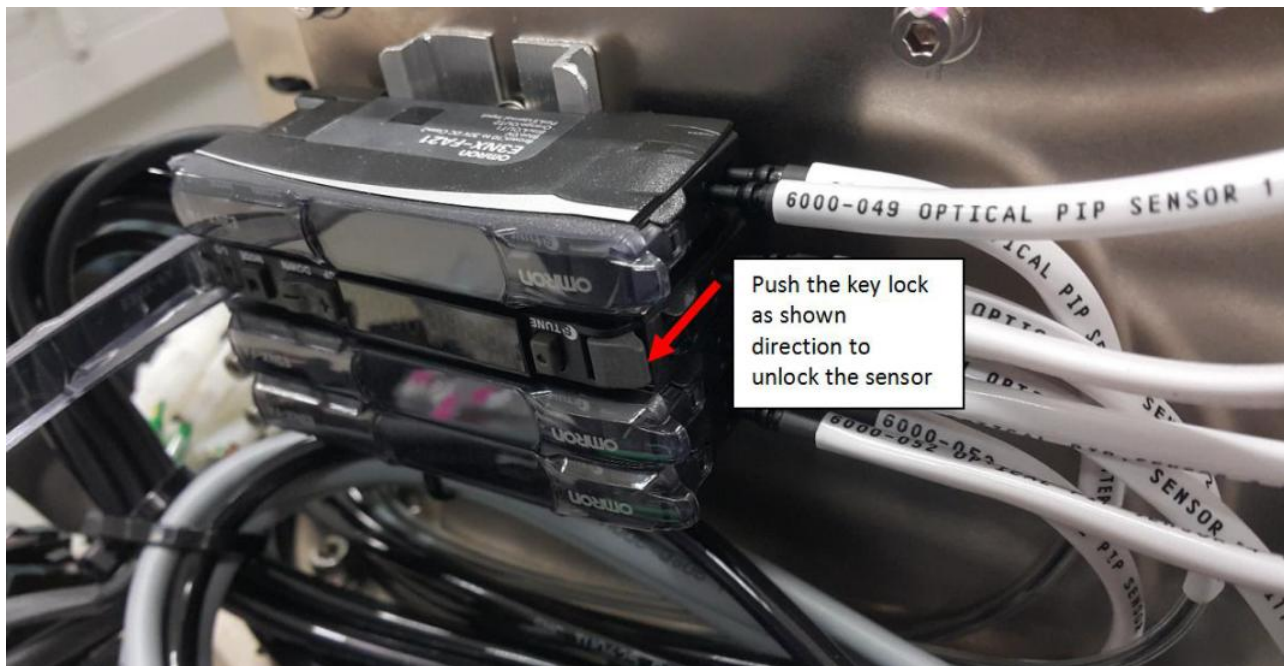


Figure 12 Sensor Amplifier

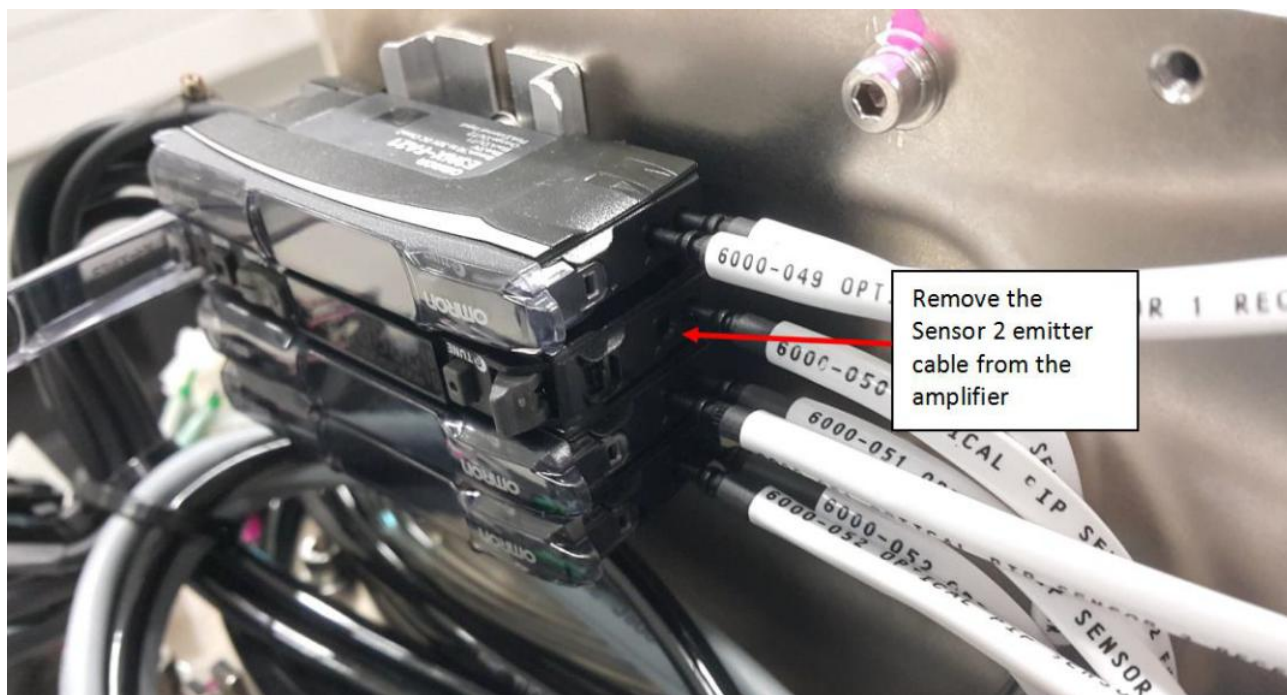


Figure 13 Sensor 2 Emitter Cable

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

II. Optical PIP Sensor 2 Receiver Removal

1. For Sensor 2 Receiver, remove BHCS M3×6 (2x) as shown in Figure 14.

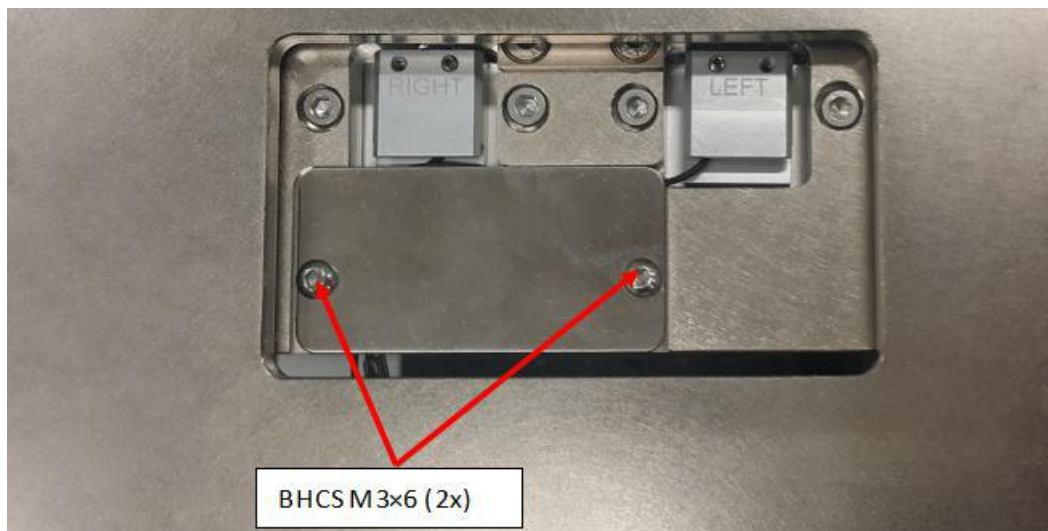


Figure 14 Sensor 2 Receiver

2. Remove SHCS M3×6 (2x) from Sensor 2 Receiver as shown in Figure 15. Loosen SHCS M2.5×3 as shown in Figure 15. Remove Optical Sensor 2 Receiver Head from pip sensor mounting plate as shown in Figure 16

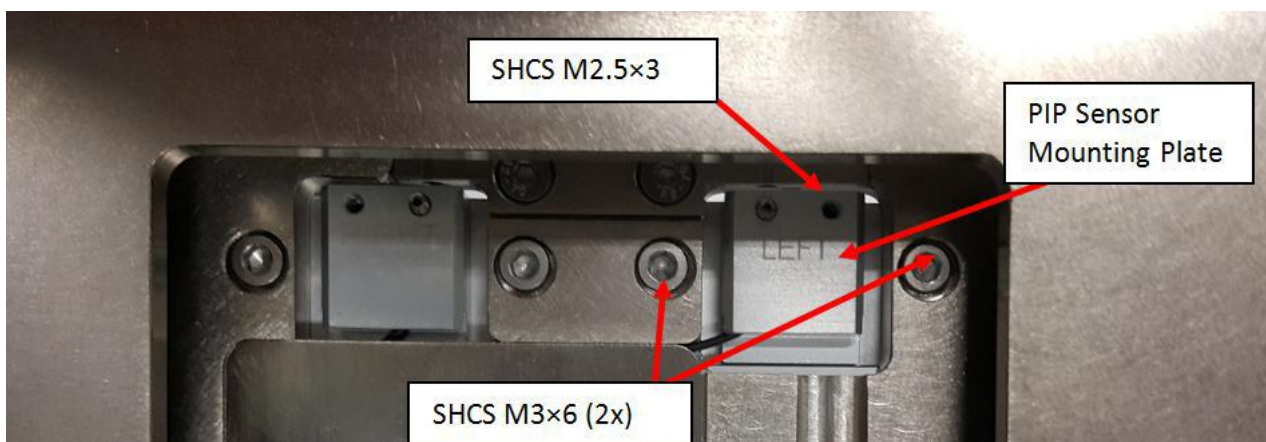


Figure 15 Sensor 2 Receiver

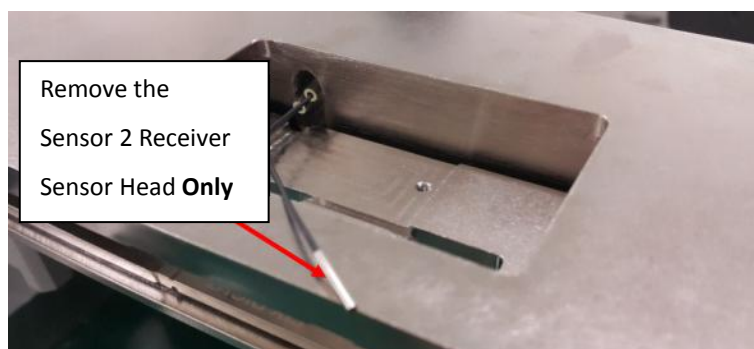


Figure 16 Sensor 2 Receiver

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

3. Remove Sensor 2 Receiver Cable by refer to below route as Figure 17 to Figure 20.



Figure 17 Sensor 2 Receiver Cable Routing

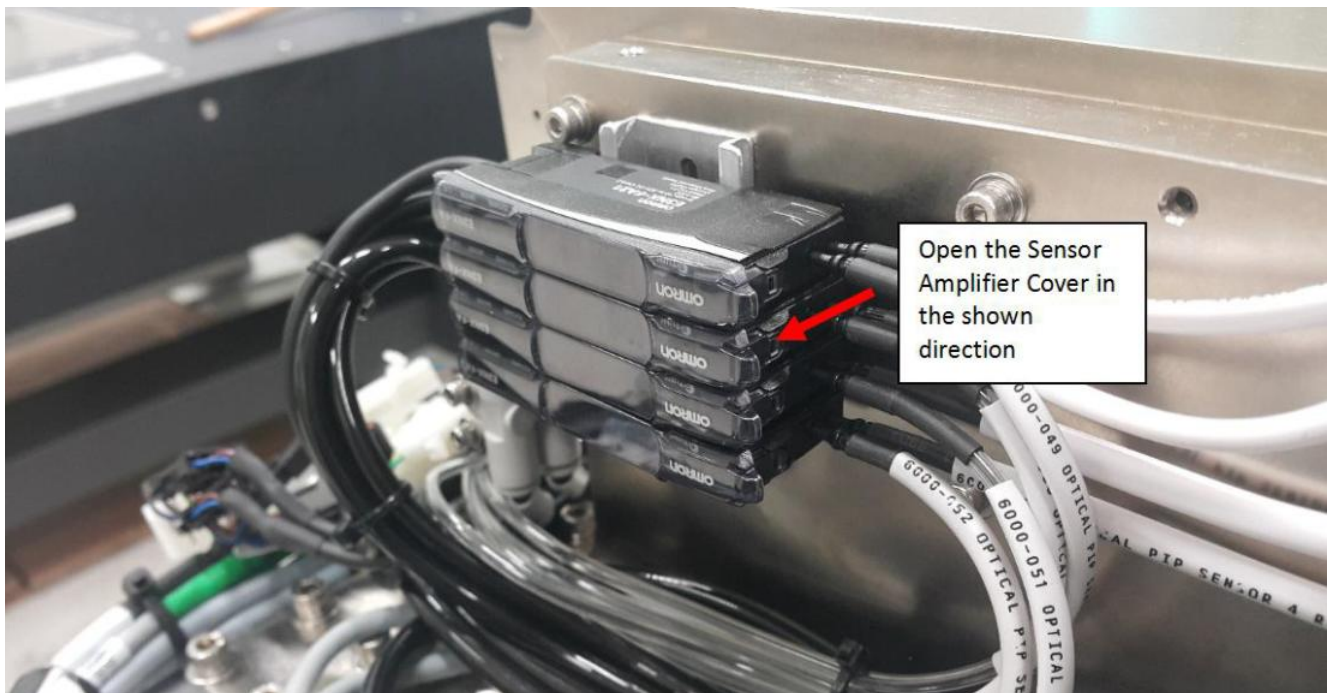


Figure 18 Sensor Amplifier Cover

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017



Figure 19 Sensor Amplifier



Figure 20 Sensor 2 Receiver (Amplifier)

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

B. Installation of the Optical PIP Sensor 2 (Part ID: 4XASP-A151 / 2000-0016)

I. Installation of the Sensor 2 Emitter

1. Install the new (4XASP-A151 / 2000-0016) S2EX Left Slow PIP Cable as shown below in Figure 21.

Remarks:Emitter and receiver are labeled accordingly.

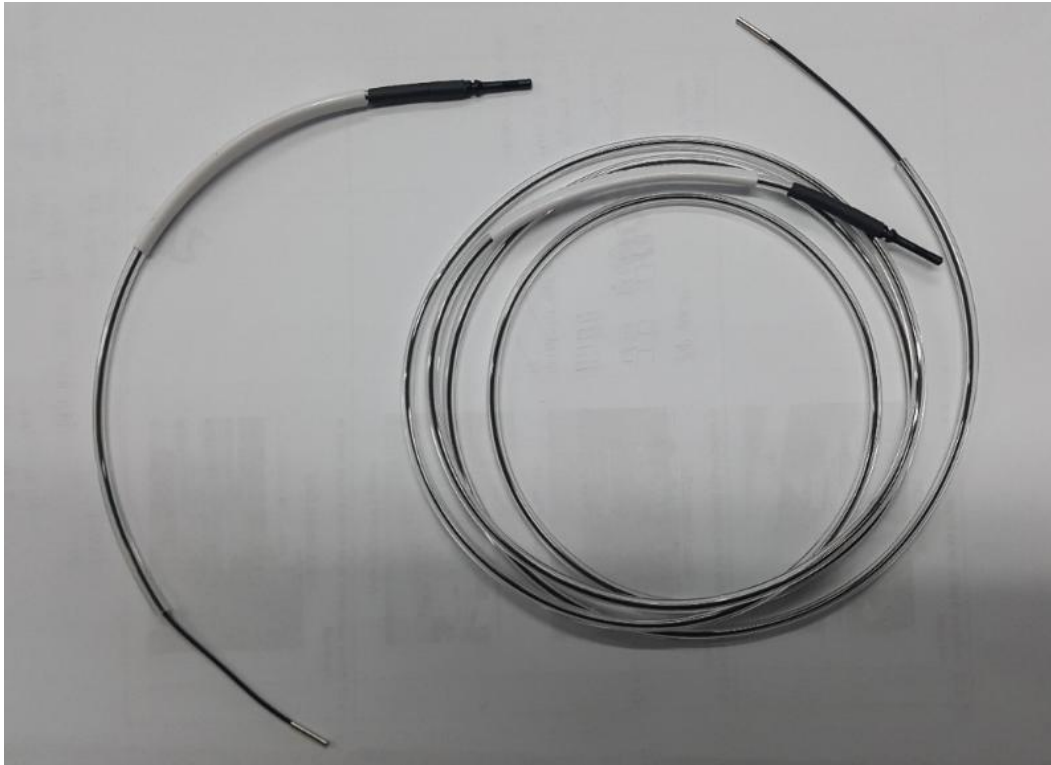


Figure 21 4XASP-A151 / 2000-0016 Left Slow PIP Cable

2. For emitter installation, insert Sensor 2 Emitter Head into 4JAXY-058 (PIP Sensor Mounting Plate 2) as shown from Figure 22 to Figure 25

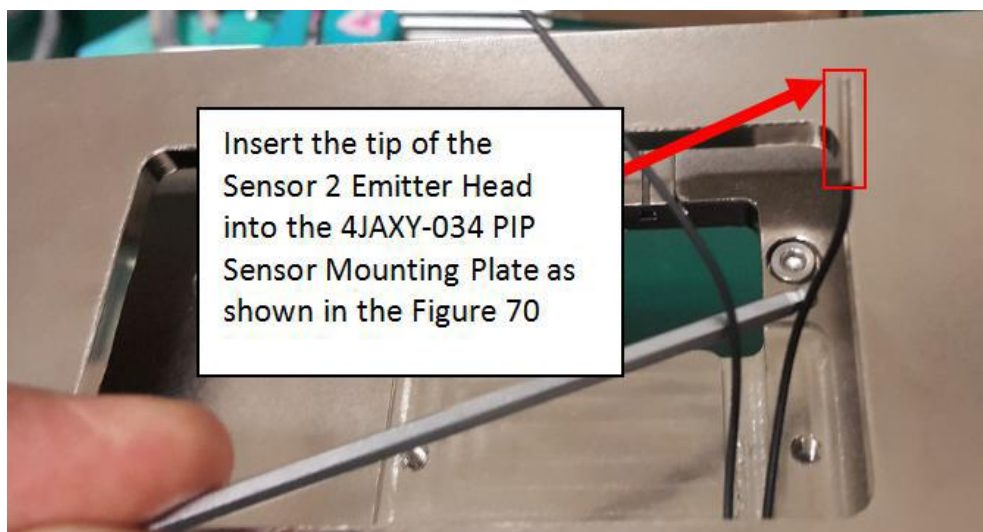


Figure 22 Sensor 2 Emitter Head

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

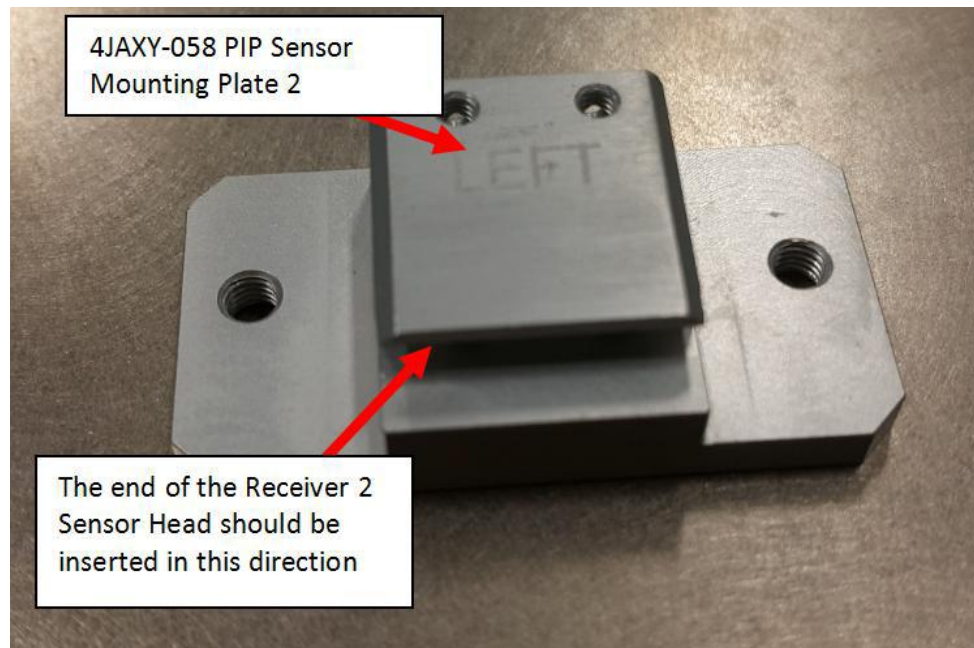


Figure 23 4JAXY-058 PIP Sensor Mounting Plate

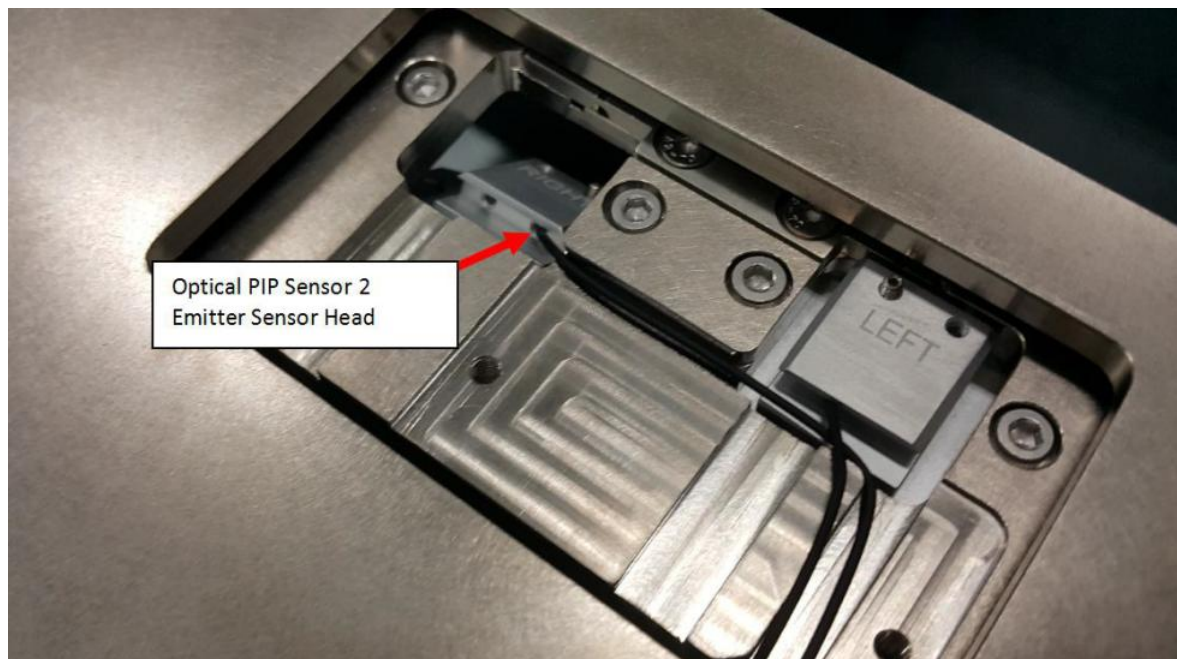


Figure 24 Optical PIP Sensor 2 Emitter Head

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

3. Push 4JAXY-058 PIP Sensor Mounting Plate 2 to maximum direction (Blue arrow) and align with the line as shown in the Figure 25.

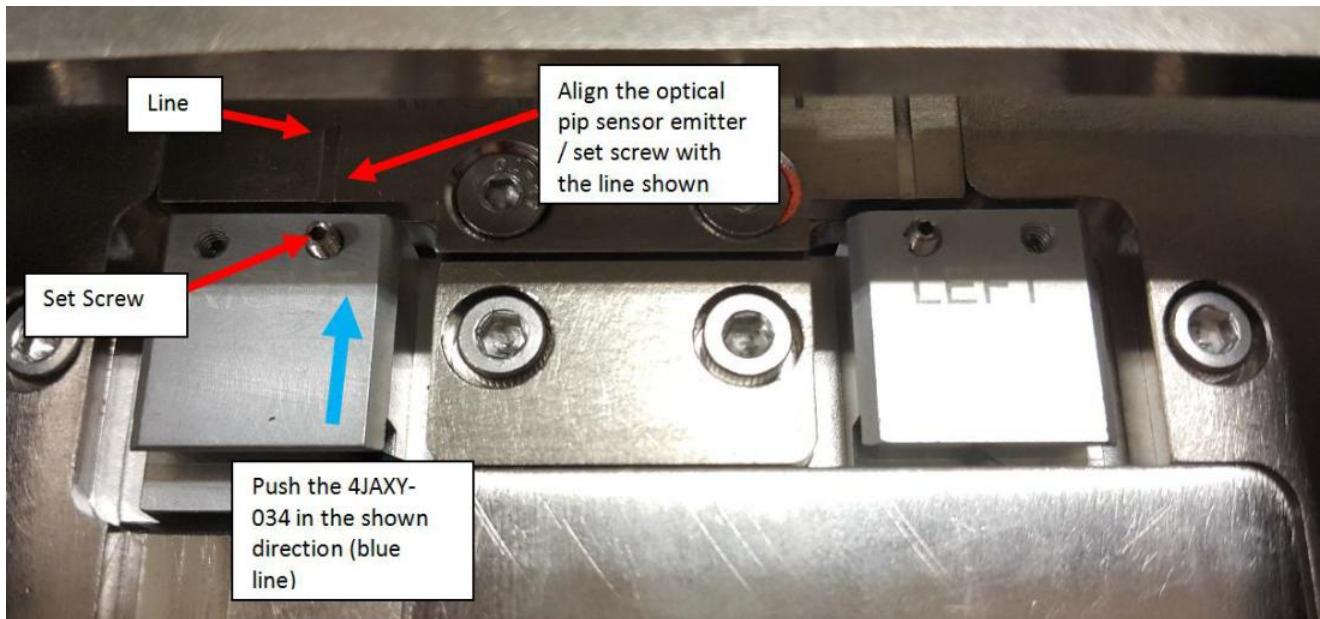


Figure 25 4JAXY-058

4. Tighten SHCS M3×6 (2x) as shown in Figure 26.

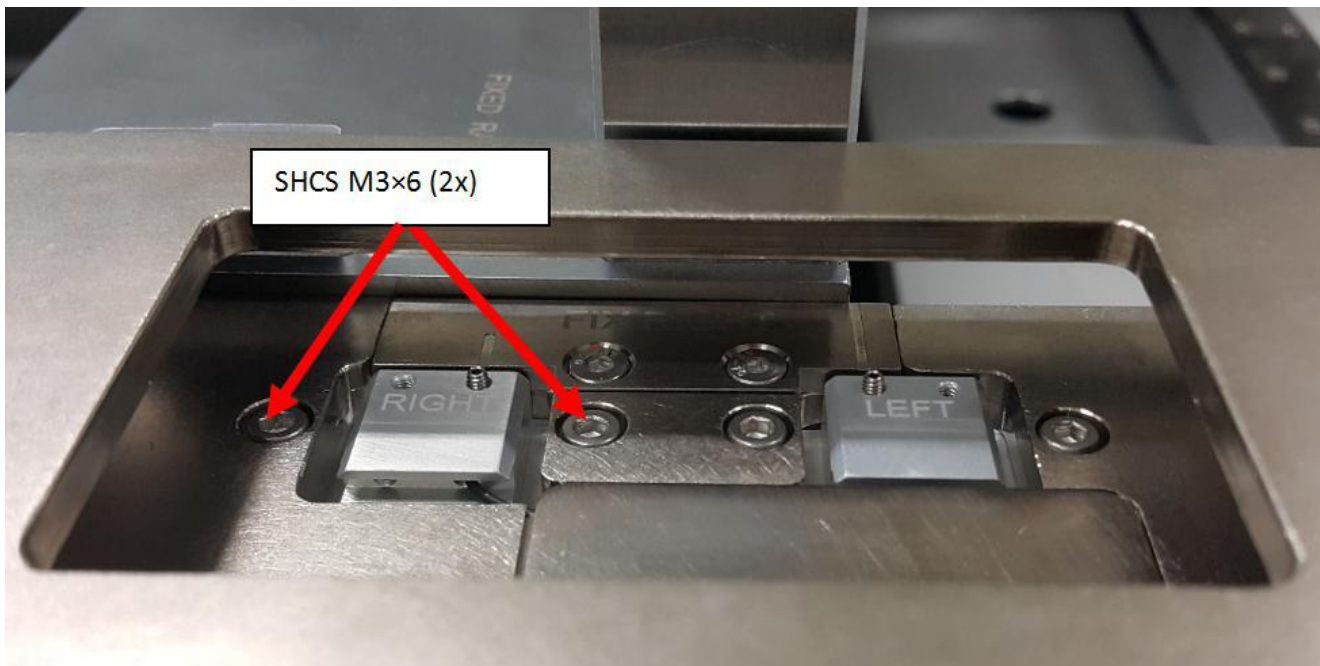


Figure 26 SHCS M3x6 (2x)

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

5. Ensure Sensor emitter 2 head do not protrude from 4JAXY-058 PIP Sensor Mounting Plate 2 as shown in Figure 27
6. Tighten back Set Screw M2.5x3 as shown in Figure 27

Remarks: Caution on Set Screw Fastening



Figure 27 4JAXY-058

7. The routing for Optical Sensor 2 Emitter is as shown below in Figure 28.

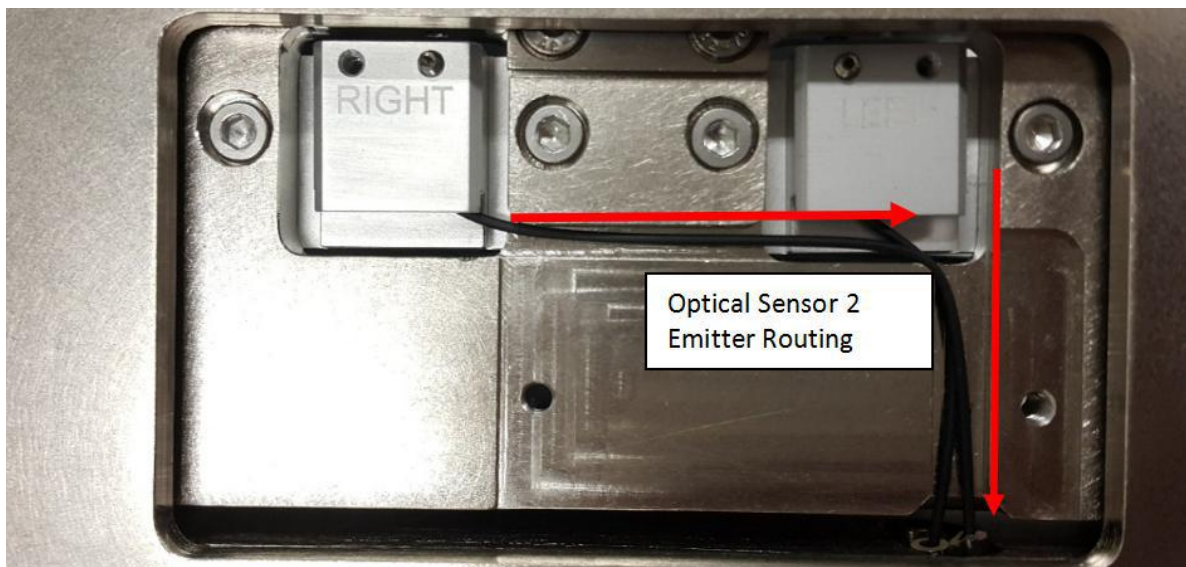


Figure 28 Optical Sensor 2 Emitter Routing

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

8. Close Optical PIP Sensor cover with BHCS M3×6 (2x) as shown in Figure 29.

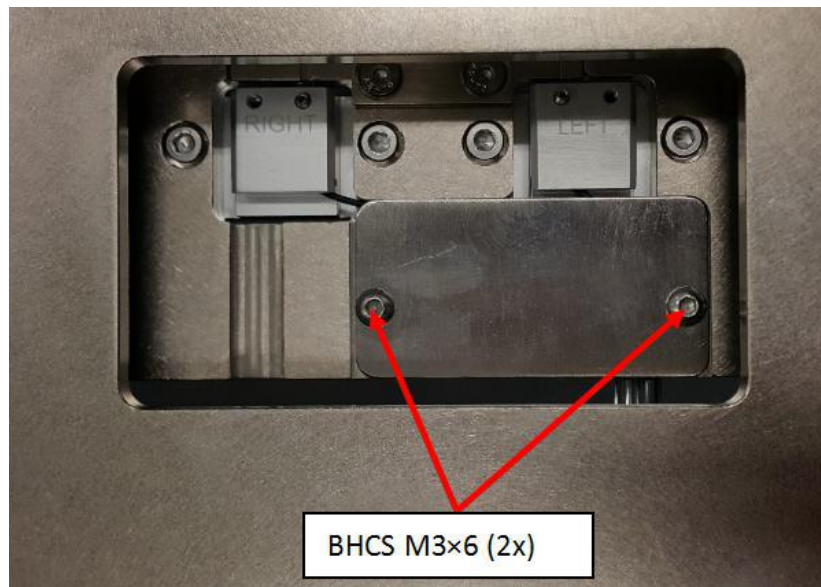


Figure 29 PIP Sensor Cover

9. Route Sensor 2 Emitter Cable by refer Figure 30 to Figure 35.

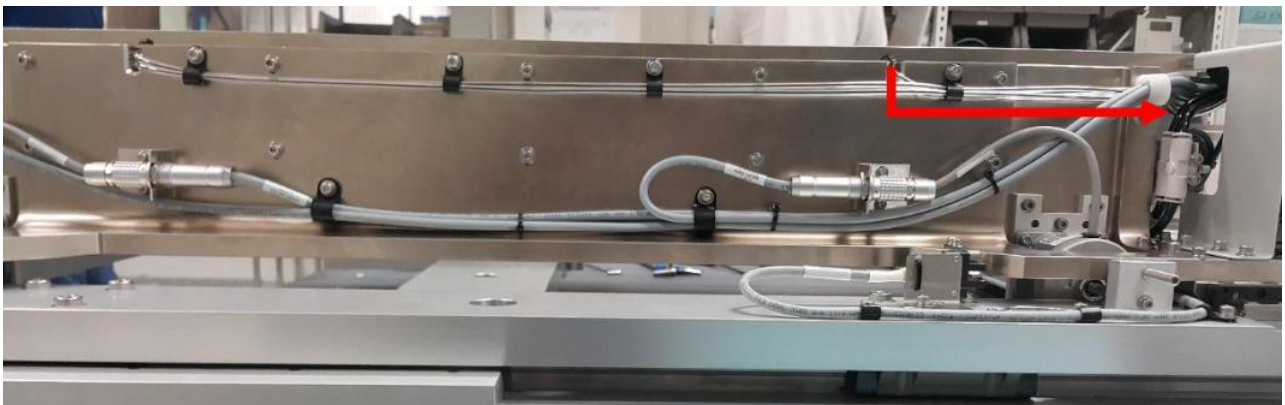


Figure 30 Sensor 2 Emitter Cable Routing

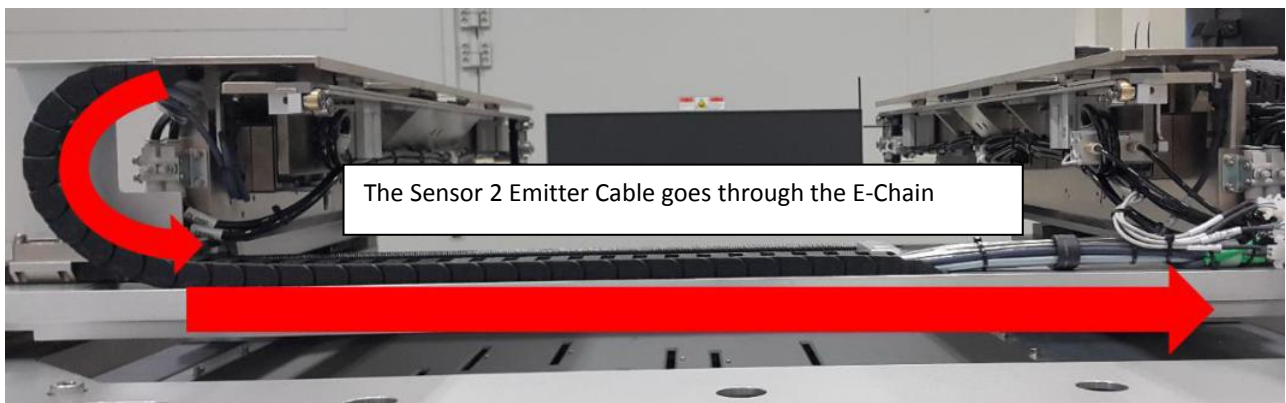


Figure 31 Sensor 2 Emitter (AWA E-Chain)

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

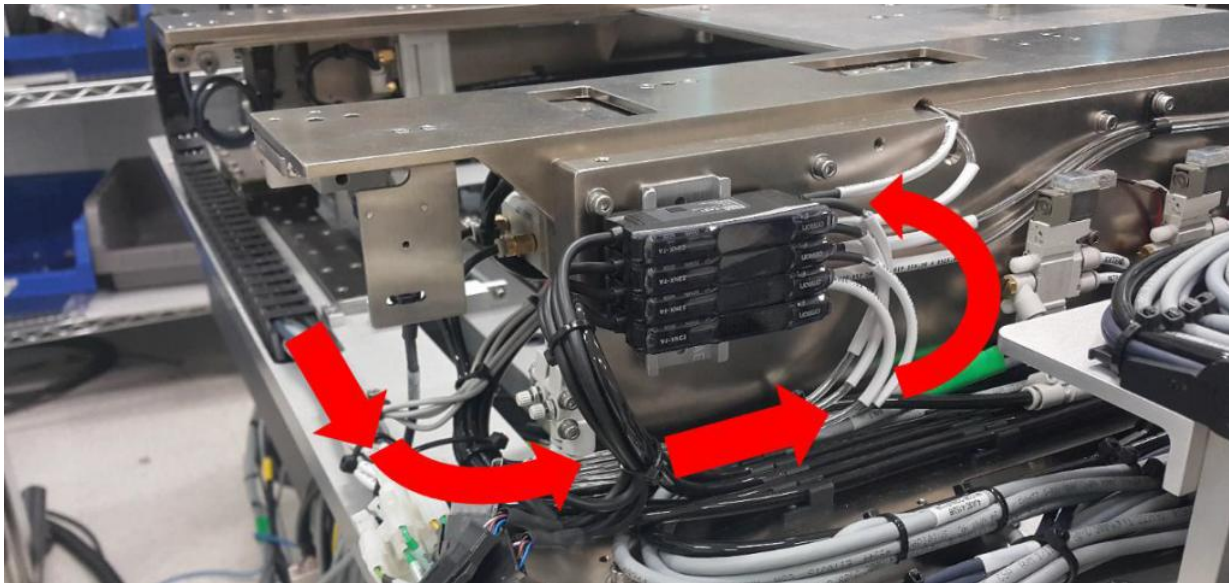


Figure 32 Sensor 2 Emitter (Sensor Amplifier)



Figure 33 Sensor 2 Emitter Cable

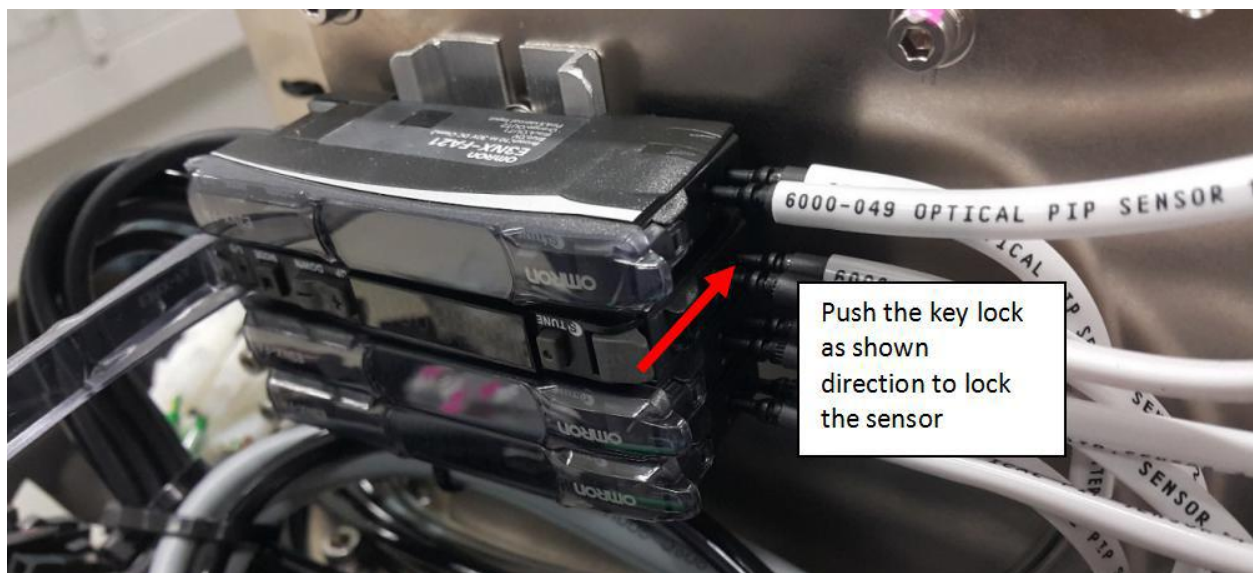


Figure 34 Sensor Amplifier

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

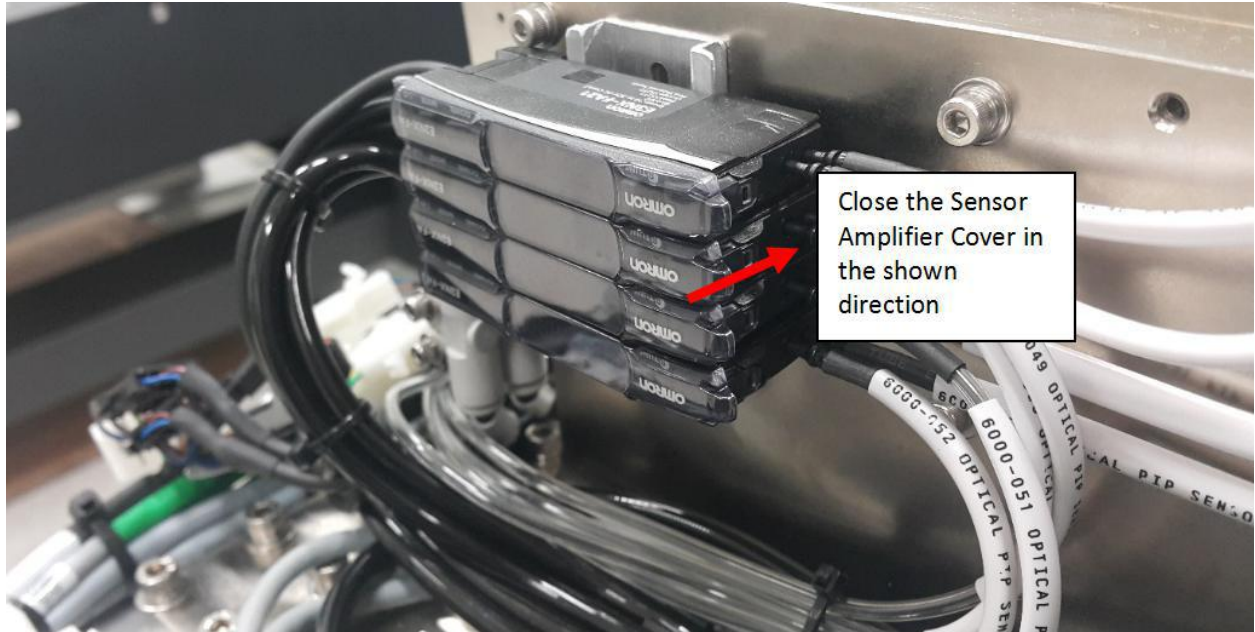


Figure 35 Sensor Amplifier Cover

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

II. Installation of the Sensor 2 Receiver

1. For Sensor 2 Receiver, insert the tip of Sensor 2 Receiver into 4JAXY-034 PIP Sensor Mounting Plate as shown below in Figure 36.

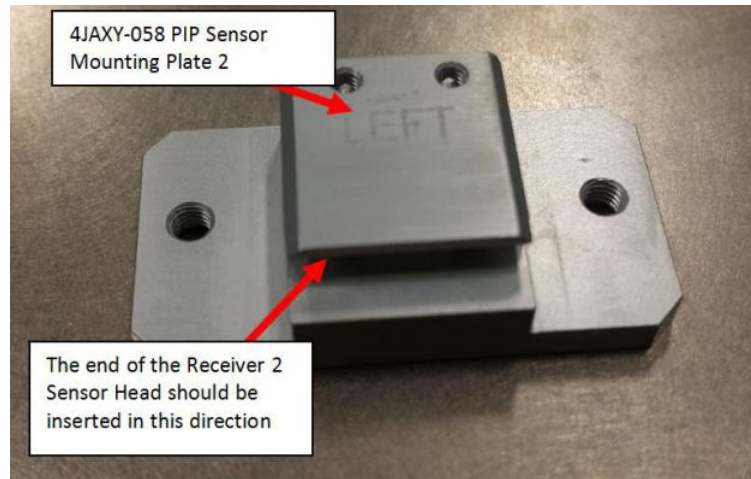


Figure 36 Sensor 2 Receiver Head

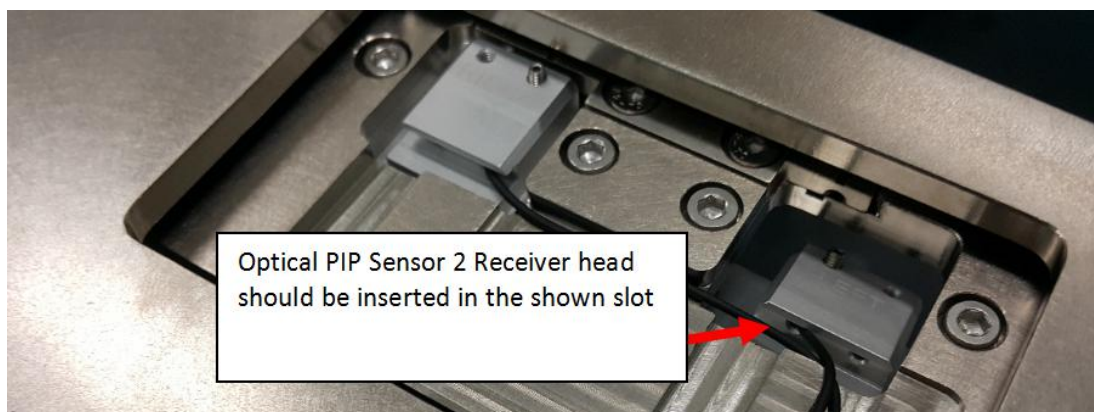


Figure 37 Sensor 1 Receiver

2. Push 4JAXY-034 PIP Sensor Mounting Plate to maximum direction (Blue arrow) and align with the line as shown in Figure 39

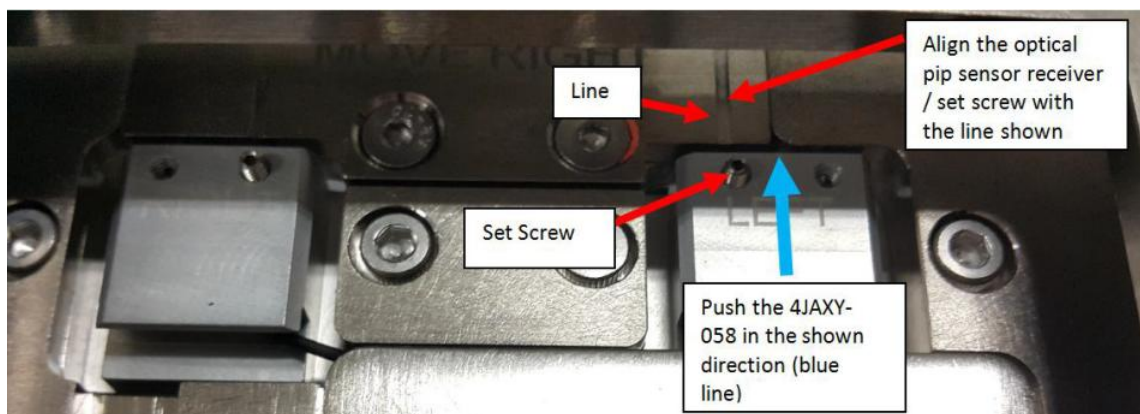


Figure 38 4JAXY-034

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

3. Tighten SHCS M3×6 (2x) as shown in Figure 39

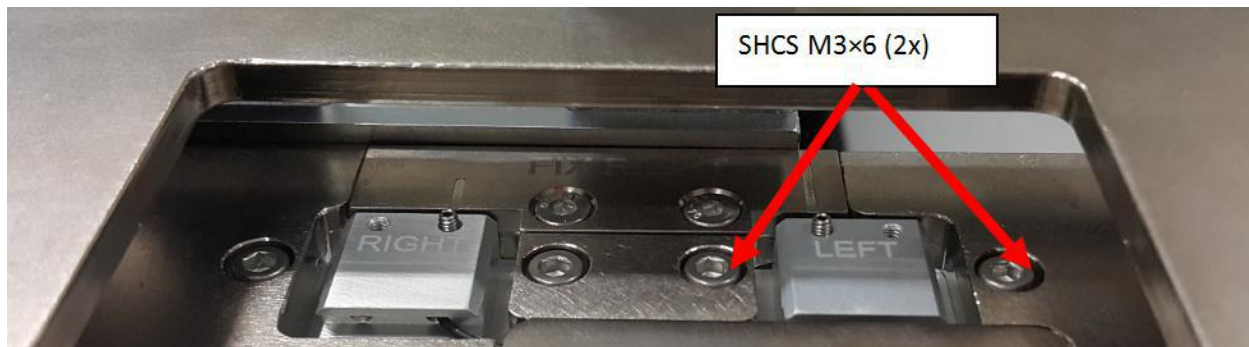


Figure 39 SHCS M3×6 (2x)

4. Ensure Sensor 2 Receiver Head do not protrude from 4JAXY-034 PIP Sensor Mounting Plate as shown in Figure 40.
5. Tighten Set Screw M2.5×3 as shown in Figure 40.

Remarks: Caution on Set Screw Fastening



Figure 40 4JAXY-058

6. The routing for Optical PIP Sensor 2 Receiver is as shown in Figure 41.

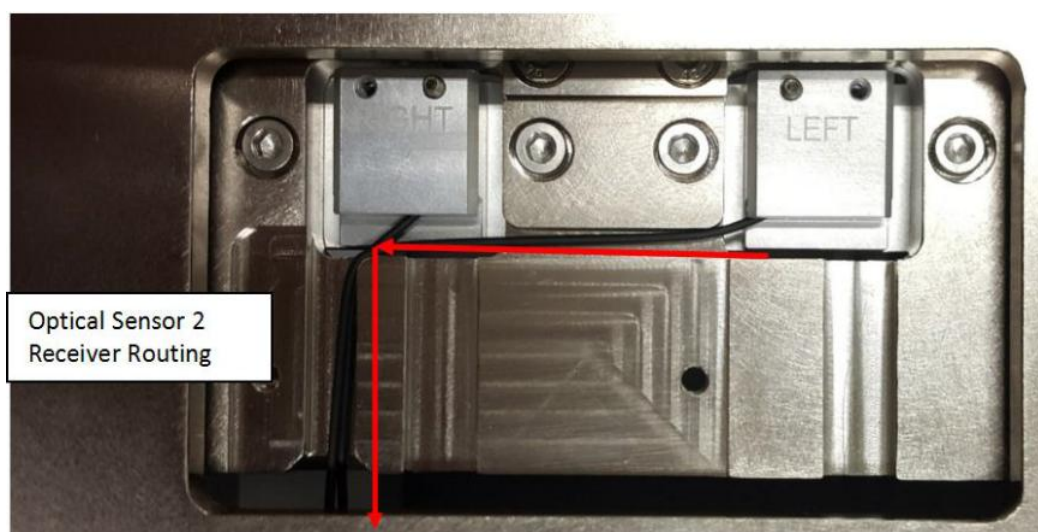


Figure 41 Optical PIP Sensor 2 Routing

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

7. Close Optical PIP Sensor Cover with BHCS M3×6 (2x) as shown in Figure 42.

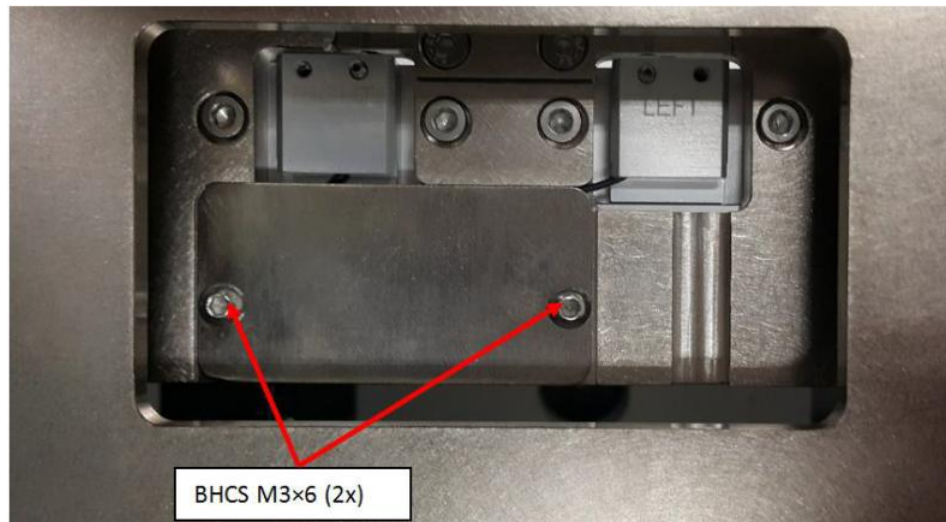


Figure 42 BHCS M3×6 (2x)

8. Route Sensor 2 Receiver cable by refer Figure 43 to Figure 47.



Figure 43 Sensor 2 Receiver Cable Routing



Figure 44 Sensor 2 Receiver Routing

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Optical PIP Sensor Replacement Procedure (Sensor 2,2000-0016)	Effective Date	16 Mar 2017

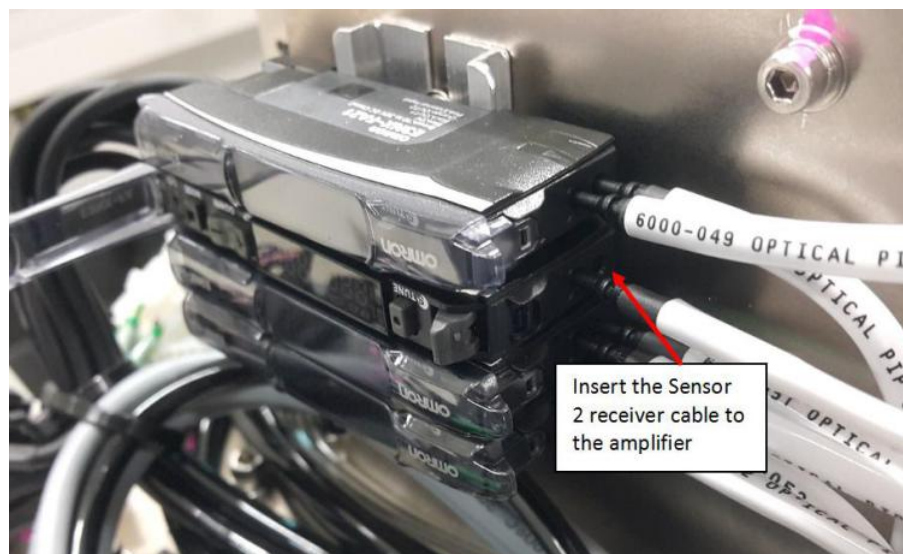


Figure 45 Sensor 2 Receiver Cable



Figure 46 Sensor 2 Amplifier

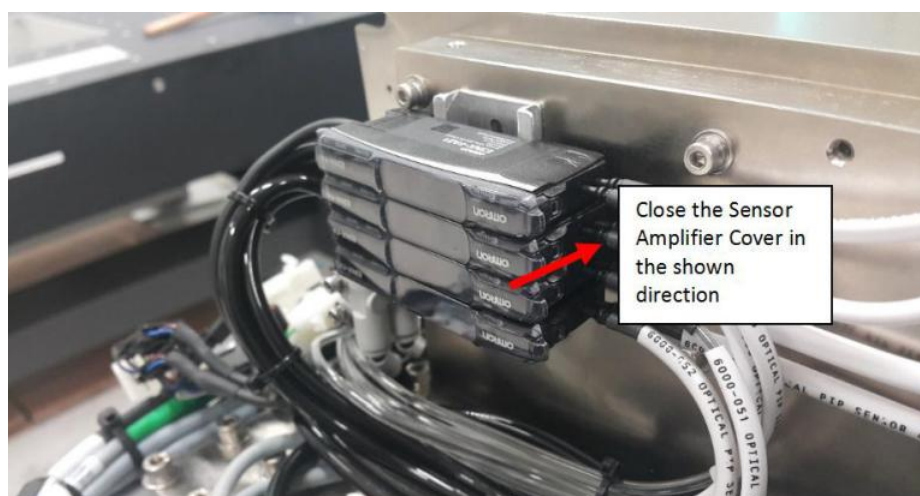


Figure 47 Sensor 2 Amplifier Cover